

Challenge 9: Union-Closed Sets Conjecture

Definition 1 (Union-closed set). A set $\mathcal{F}$ of sets is *union-closed* if for any two sets in $\mathcal{F}$, their union is also in $\mathcal{F}$. For a nonempty finite set $\mathcal{F}$, the *density* of an element x with respect to $\mathcal{F}$ is defined to be

$$\frac{|\{F \in \mathcal{F} : x \in F\}|}{|\mathcal{F}|}$$

Conjecture 1.1 (Union-closed sets conjecture [1]). For every finite union-closed set $\mathcal{F} \notin \{\emptyset, \{\emptyset\}\}$, there exists an element x of density at least one half.

We are interested in the following parametrized version of the conjecture.

Conjecture 1.2 (Scaled union-closed sets conjecture). Let $r \in \mathbb{N}_{>2}$. For every finite union-closed set $\mathcal{F} \notin \{\emptyset, \{\emptyset\}\}$, there exists an element x of density at least $\frac{1}{2} - \frac{1}{r}$.

The above conjecture is open for all values of $r > 2$.

References

- [1] N. Nagel. *Notes on the Union Closed Sets Conjecture*. 2023. arXiv: 2208.03803 [math.CO].
URL: <https://arxiv.org/abs/2208.03803>.